Begum Sultan Jahan Hall

Aligarh Muslim University

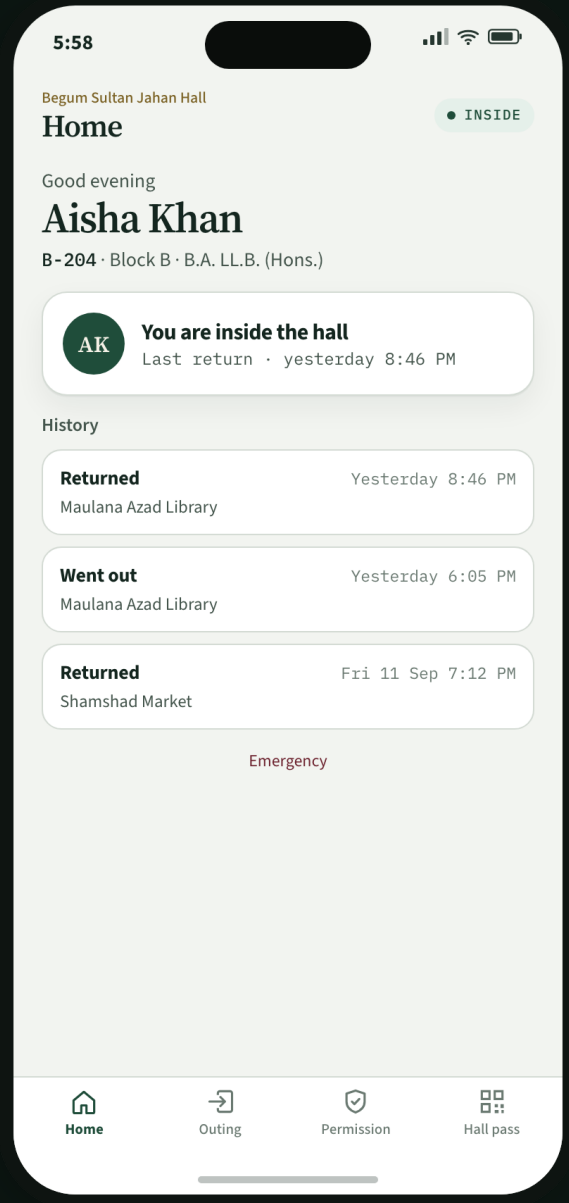
Hostel Management App Proposal

Built for

Resident | Warden | Gate staff | Provost

Try a demo

bsj-hostel-demo.vercel.app



The case

Today

Movement is a paper register at the gate.

With the app

- Students and admins check in and check out from one app.
- Residents request special permission.
- Wardens approve special permission.
- Parents are notified if a resident is late.

How the hall pass works

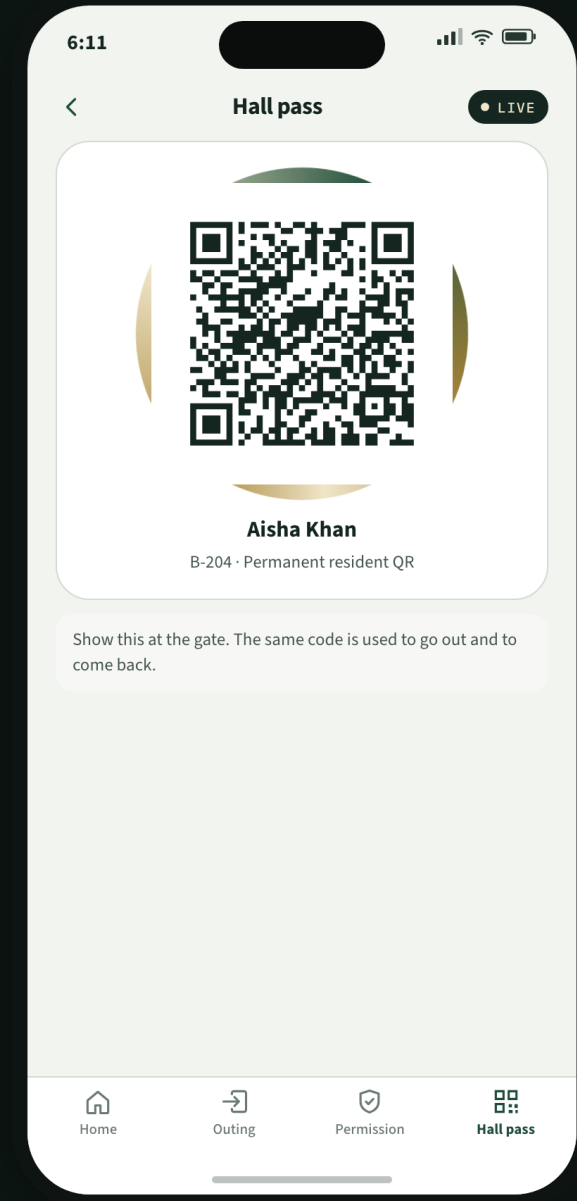
Every resident has a hall pass on the phone: a QR code. At the gate, staff scan this hall pass. The app already knows whether that resident is inside the hall or outside.

If the resident is inside the hall

The resident shows the hall pass. Gate staff scan it. The app records that the resident has gone out. Destination and return time were already set in the app.

If the resident is outside

The resident shows the same hall pass. Gate staff scan it. The app records that the resident has come back. Nothing extra to fill in.



Built for everyone

Resident

Plans an outing, requests special permission, and shows the hall pass at the gate.

Warden

Sees who is inside, outside, or late. Approves or rejects special permission.

Gate staff

Scans the resident hall pass. The app records exit or return.

Provost

Sees the whole hall. Can export the register.

Scenario 1. Normal outing

Resident

1. Sets destination and return time.

Resident

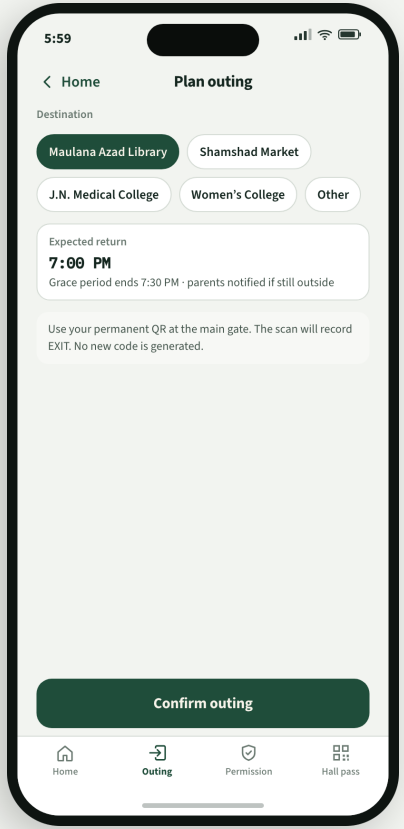
2. Opens the hall pass at the gate.

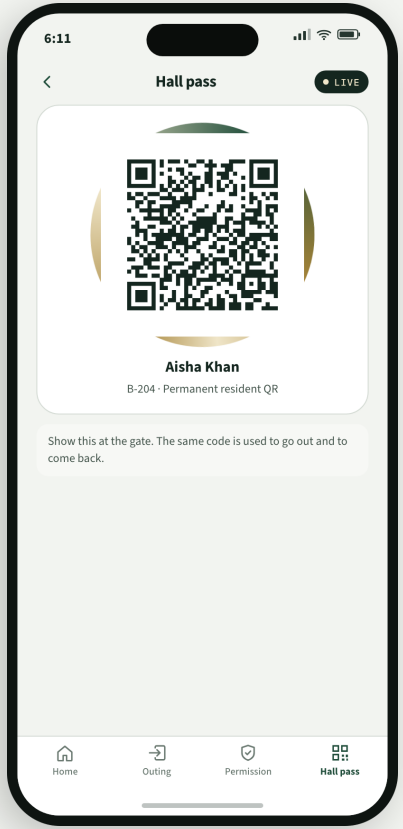
Gate staff

3. Scans the hall pass.

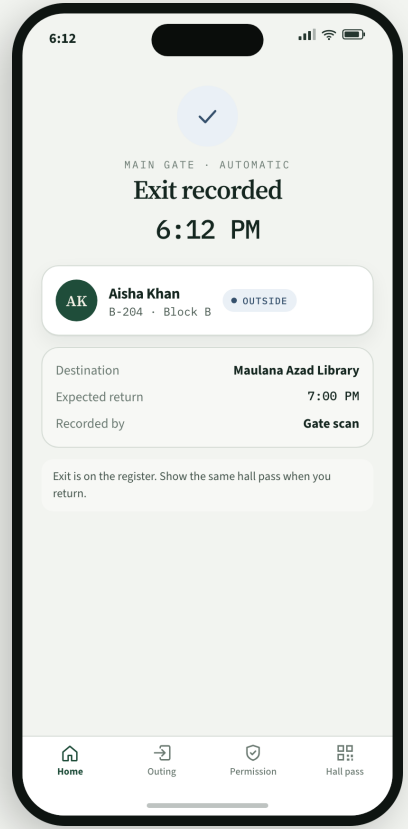
Resident

4. Exit is recorded.









Scenario 2. A resident comes back

Resident

1. Opens the same hall pass.

Gate staff

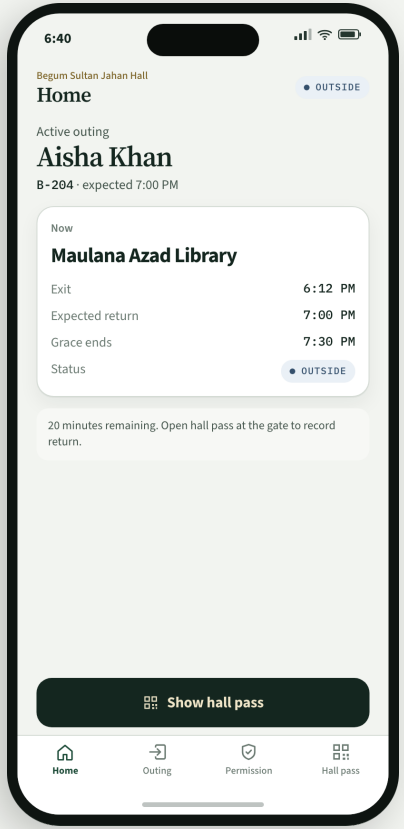
2. Scans it. Return is recorded.

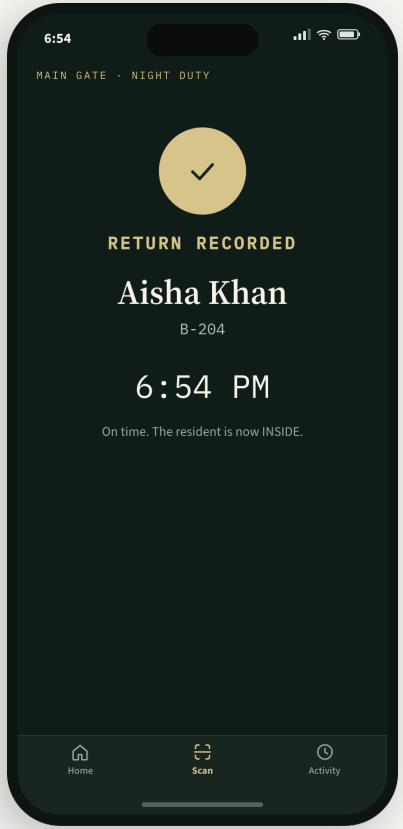
Warden

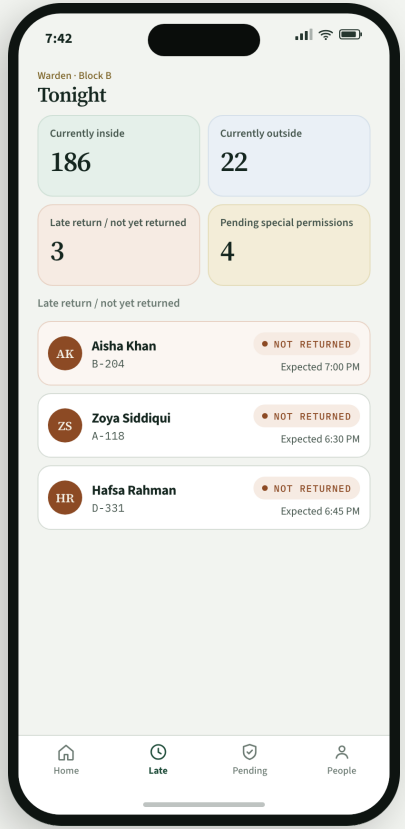
3. If late, the warden sees the late list.

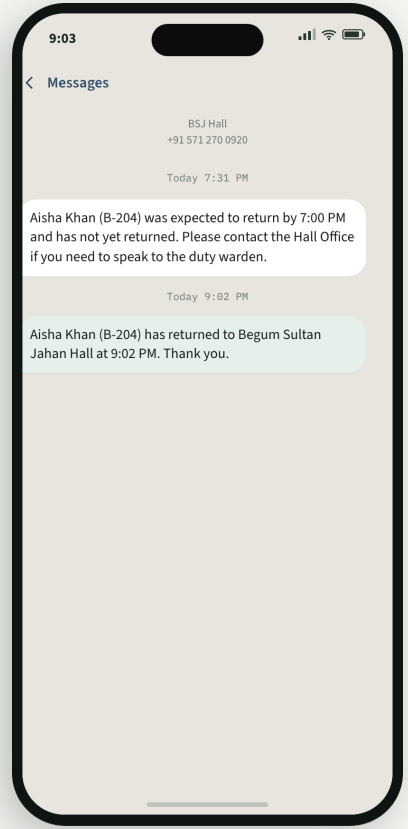
Parent phone

4. A parent SMS is sent, then another on return.









Scenario 3. Special permission

Resident

1. Chooses late-night or home going.

Resident

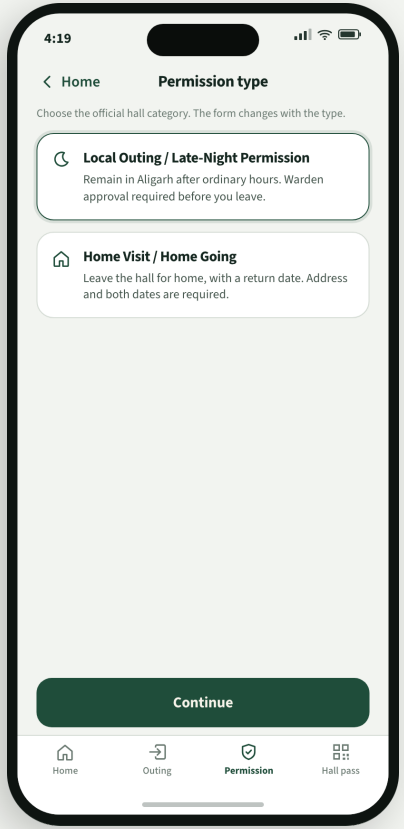
2. Fills the form and sends it to the warden.

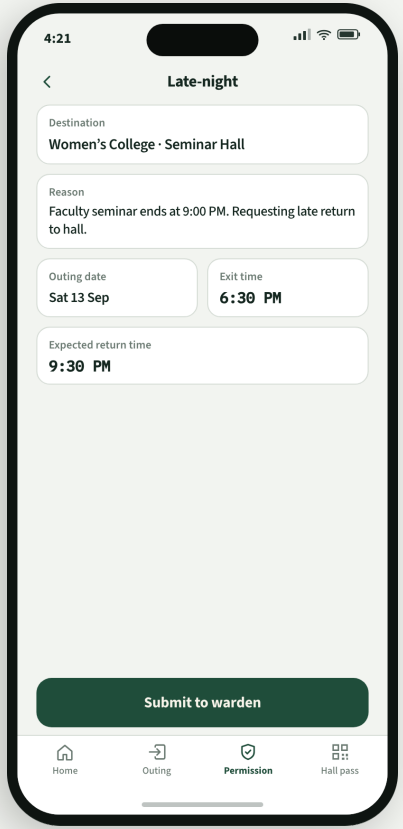
Warden

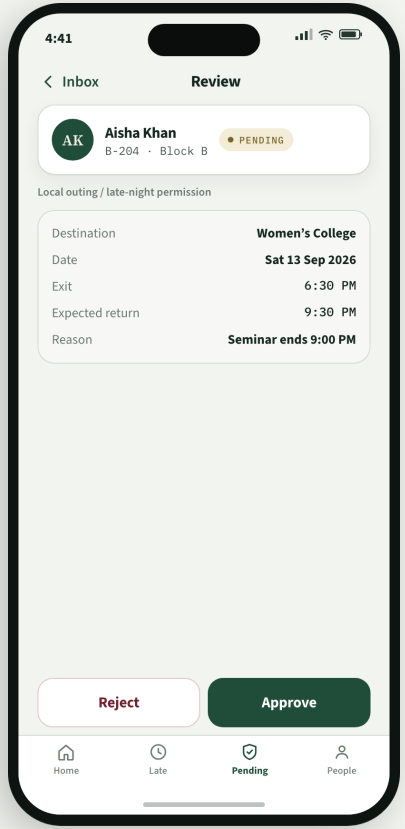
3. Approves or rejects.

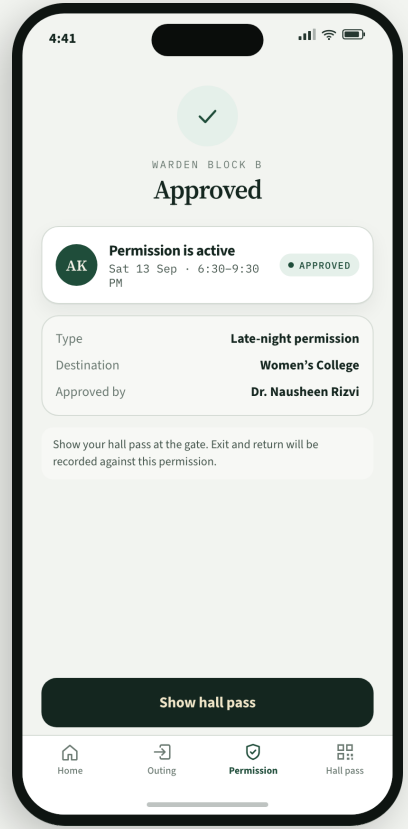
Resident

4. If approved, the same hall pass is used at the gate.









https://bsj-hostel-demo.vercel.app/permission

Scenario 4. Gate staff

Gate staff

1. Opens Scan QR.

Gate staff

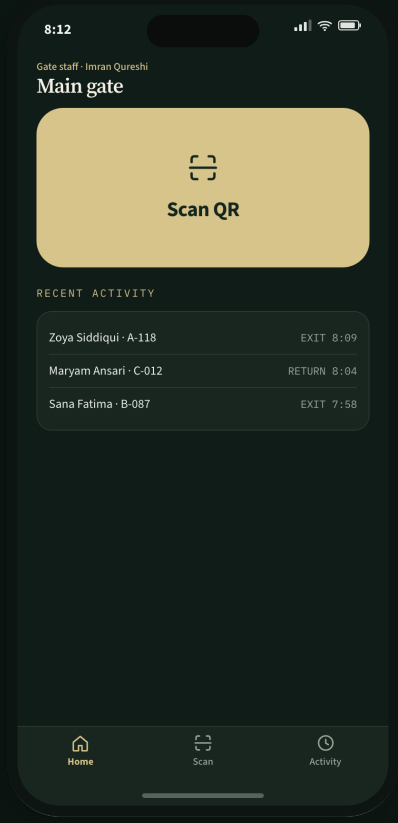
2. Scans the resident hall pass.

Gate staff

3. Shows Exit recorded or Return recorded.

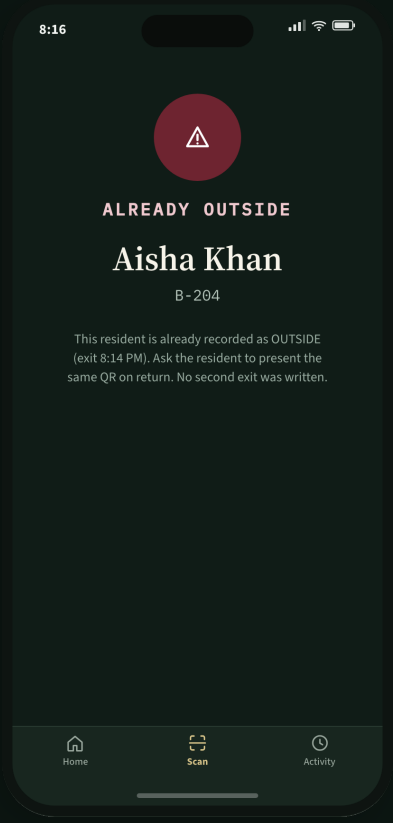
Gate staff

4. If the scan cannot be used, the screen says what to do.









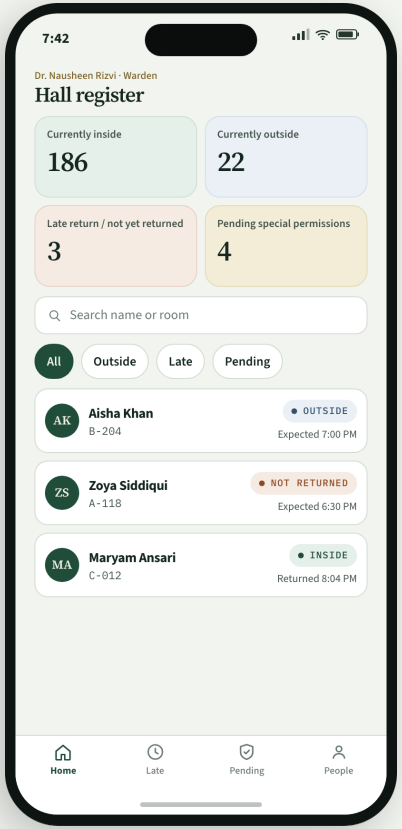
Scenario 5. Warden

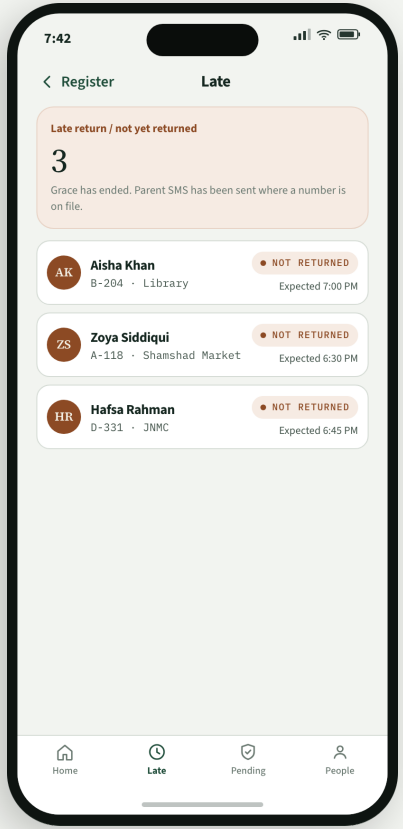
Warden
1. Opens the hall register.

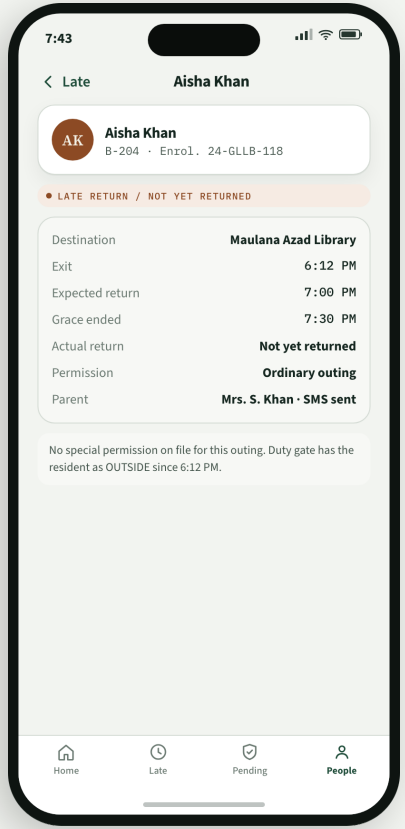
Warden
2. Opens the late list.

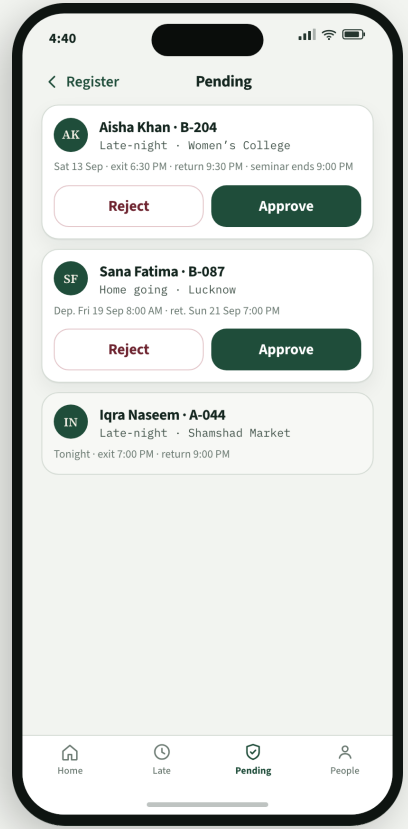
Warden
3. Opens a resident record.

Warden
4. Approves or rejects special permission.







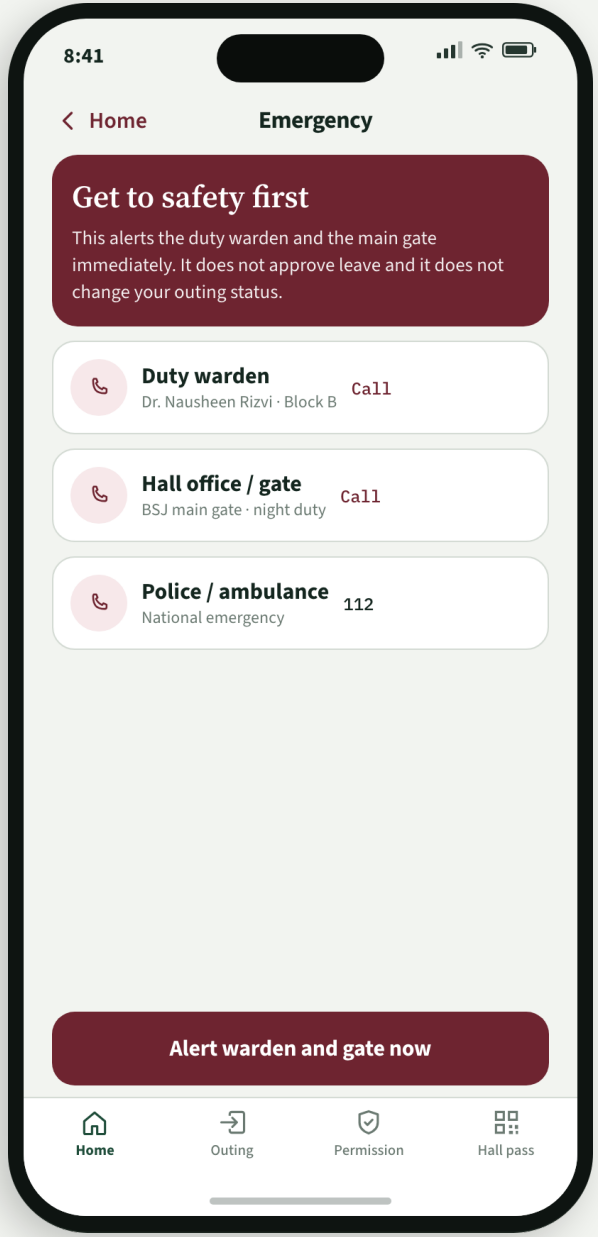


Optional

Emergency

A resident can alert the duty warden and the gate, and call from the phone. This is not a leave form.

Skip this if the hall does not want it in the first version.



What we will build

1. Normal outing: destination, return time, scan at the gate. No special permission.
2. Return, including late return and parent SMS
3. Special permission: late-night and home going
4. Gate scanning
5. Warden register: inside, outside, late, pending

Optional: emergency contacts on the resident phone.

Running cost (estimate)

Estimate for 400 residents and 10 admins. iPhone and Android app. Developer cost to be confirmed.

Item	Amount
Developer cost, one-time	To be confirmed
India VPS, 2 GB, about ₹500 a month	₹6,000 each year
Parent SMS, late returns only	₹2,000 each year
App Store, Apple Developer $99 a year	₹8,500 each year
Play Store, Google $25 once	₹2,100 in year 1
Estimated total with Apple and Google accounts, year 1	**₹18,600**
Estimated total with Apple and Google accounts, each year after	**₹16,500**

With personal Apple and Google accounts, same as the ICACON Companion app, App Store and Play Store are ₹0. Running cost is then ₹8,000 a year.

India VPS: one 2 GB server in an Indian data centre (Mumbai or Bangalore) is about ₹500 a month, so ₹6,000 a year. It runs the app backend, the database, and a nightly backup. 400 residents and 10 admins fit on this size.

Parent SMS: only if a resident is late. Two texts: one when grace ends, one when they return. About 8 late returns a night × 250 term days × 2 texts ≈ 4,000 SMS a year. Indian transactional SMS is about ₹0.20 each, about ₹800. We estimate ₹2,000 so a busier year still fits.